

$EL_{P0} = 7.75 \text{ MeV}$

8(a)

The X-ray tube is evacuated so that the electrons can move from the filament to the heavy metal target without colliding with air molecules and getting scattered.

8(b)

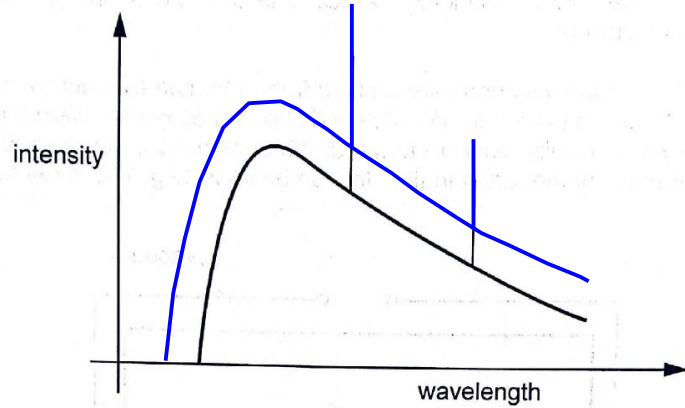
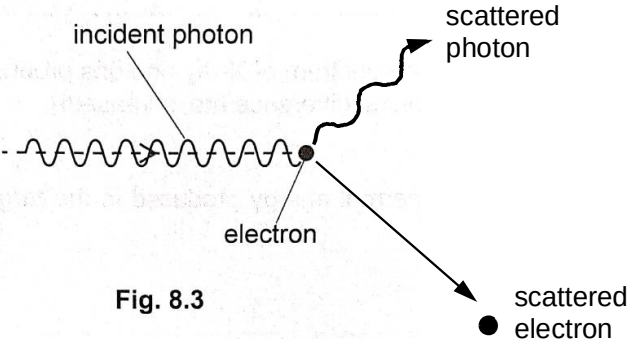


Fig. 8.2

No.	Solution
	<i>Note: The 3 marking points include a lower minimum wavelength, a more intense peak at a smaller wavelength and the characteristic vertical line (K-lines) at the same wavelength.</i>
8(c)(i)	Thermal energy $E = 0.99 \times P \times t$ $= 0.99 \times V \times I \times t$ $= 0.99 \times 65 \times 10^3 \times 0.12 \times 1.1$ $= 8.494 \times 10^3$ $\approx 8.49 \times 10^3 \text{ J}$ <i>Note: A clear step-by-step approach in showing clear working is important.</i>
8(c)(ii)	Thermal energy to raise temperature = Thermal energy produced in the target $mc\Delta\theta = E$ $12 \times 10^{-3} \times 130 \times \Delta\theta = 8.494 \times 10^3$ $\Delta\theta = 5.445 \times 10^3$ $\approx 5.45 \times 10^3 \text{ }^\circ\text{C}$
8(c)(iii)	The anode is rotated to ensure the electron beam hits all parts of the anode rather than just a specific part, so that the thermal energy generated is more evenly distributed to the entire anode. This prevents excessive temperature increase which might cause the anode to melt. <i>Note: Detail is expected even though this is a 1-mark question. Insufficient detail is often given by suggesting that the spinning is to increase the distribution or spreading of the heat energy or merely to prevent the anode target from melting. It is also incorrect to suggest that the spinning of the target cools it down.</i>
8(d)(i)	 The diagram illustrates the Compton effect. An incident photon, represented by a horizontal wavy line, moves from left to right. It collides with an electron, represented by a solid black dot. After the collision, a scattered photon moves away at an angle, represented by a wavy line with an arrow. The electron is also scattered, moving away at a different angle, represented by a straight line with an arrow. Labels with leader lines identify the 'incident photon', 'electron', 'scattered photon', and 'scattered electron'. The caption 'Fig. 8.3' is centered below the diagram. <i>Note: Labelling of the paths drawn is important.</i>
8(d)(ii)	When the X-ray photon is scattered, the photon loses energy. Since the energy of a photon is inversely proportional to its wavelength, the wavelength of the scattered photon increases.

No.	Solution
8(d)(iii)	Compton scattering decreases the level contrast of an X-ray image. The scatter radiation adds noise to the image that decreases the image sharpness and degrades the diagnostic quality of the image.
8(e)	Since attenuation $\propto Z^3$, ratio $\frac{\text{attenuation of X-rays in bone}}{\text{attenuation of X-rays in soft tissue}}$ $= \left(\frac{14}{7}\right)^3$ $= 2^3$ $= 8.0$
8(f)	Using $A = A_0 \exp(-\lambda t)$, transmitted intensity I_T and incident intensity I_0 of X-ray on matter are related by $I_T = I_0 \exp(-\mu x)$ $0.5I_0 = I_0 \exp(-\mu x_{1/2})$ $\exp(-\mu x_{1/2}) = \frac{1}{2}$ $\therefore x_{1/2} = \frac{\ln(2)}{\mu}$ $= \frac{0.693}{\mu}$
8(g)	The use of more sensitive detectors allows a lower dose of X-ray radiation for a given picture quality than film or fewer X-rays needed, both of which reduces the radiation risk to patients. <i>Note: The detail for the advantage to the patient needs to be clearly given, as in the text of the question.</i>
8(h)	The solution containing barium ($Z = 56$) results in a high attenuation of the incident intensity of X-ray. This results in a greater contrast between the transmitted X-ray photon intensities in the digestive system which comprises many types of soft tissues with similar Z numbers (average $Z = 7$). The barium solution serves as a contrast medium which helps improves contrast between different areas in the digestive system being examined. <i>Note: Detail in making the comparison between the Z numbers of barium solution and the soft tissue in the human body so as to highlight the effectiveness of X-ray imaging on soft tissues.</i>
8(i)	CT scan exposes the patient to more radiation by taking multiple X-ray images. The higher level of radiation due to multiple X-ray exposures may subject the patient to harmful ionising radiation, increasing the risk of cancer.